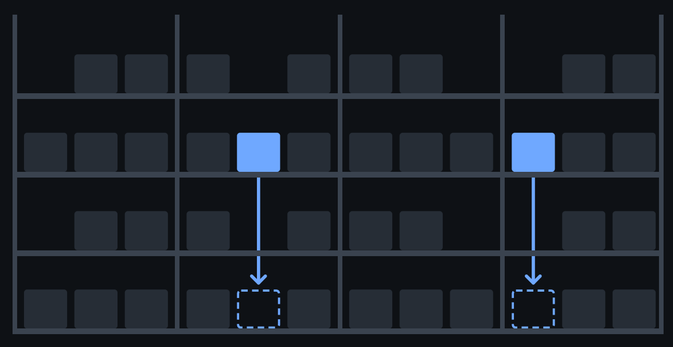
Warehouse Planning Agent

An AI agent that plans a footwear warehouse's day, every morning.



Four answers before picking starts

How much will ship?
Pairs, order lines and orders for the next 5 working days, with a likely range.

Which SKUs are in demand?
The 15 expected to sell most, and where their stock sits.

What comes down from levels 3-4?
Every box to bring within reach for the next 2 working days.

How many workers each day?
Hours of work turned into workers, and the number if the day runs high.

What makes it an agent

It is given a goal and 8 tools. It decides which tool to call next, reads the answer, checks its own plan and asks a person when it should not settle something alone.

Observe
Read the orders and grade the last plan.

Decide
Choose the forecast method that missed least, and how to staff.

Act
Forecast, find the SKUs in demand, plan the boxes and the workers.

Check
Test the plan against the roster and the racks.

Report
Write the brief and raise what a person must decide.

A real plan

Friday 20 October 2023, the first working day after the records end

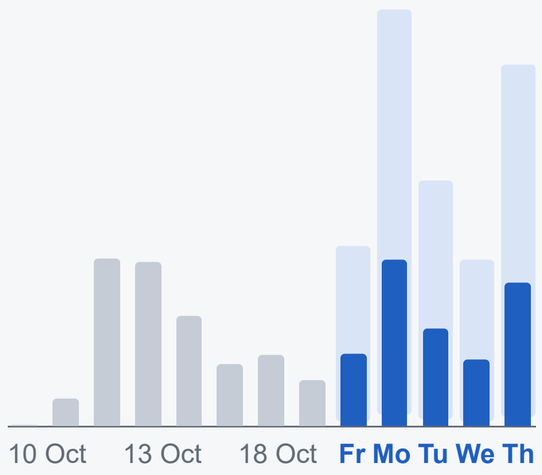
Every plan on the dashboard opens with the agent's brief:

> Friday 2023-10-20: expect about 832 pairs (54 to 2,085) in 471 order lines. Bring down 62 boxes from levels 3-4 to 1-2 before picking starts, starting with AX7BJ8 size 9 from M-24-30. Staff 1 worker, 3 if the day runs high. Rising: AKP0A7.

1. Demand

832

pairs expected, likely 54 to 2,085



2. SKUs in demand

Pairs over the next 5 working days. The top 15 take 50.5%.



3. Boxes to bring down

62

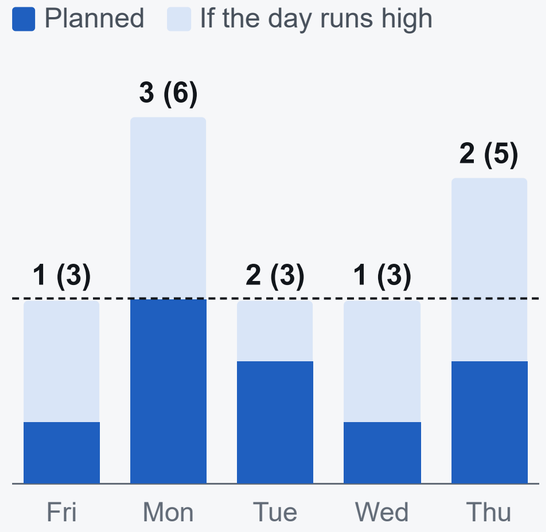
boxes from levels 3-4 to 1-2: 32 SKU sizes, 7 new pick faces

SKU size	From	To
AX7BJ8 9	M-24-30 L3	M-22-13
EUMBM7 9	Q-21-30 L3	Q-21-12
5MNTT7 11	E-14-30 L3	E-14-11

4. Workers

1

worker today, 3 if the day runs high. Dashed: the 3 on the roster.



How well it planned 134 real days

Each planned from earlier orders only, then checked against what happened. The simplest rule, the same weekday last week, misses by 75.9%.

64.9%
forecast miss, as a share of pairs shipped

79.9%
of days with enough workers planned

27.9%
of boxes brought down were needed in time

What did not work, and why

- Demand by SKU size is too lumpy to place: even knowing the next 5 days' total, sharing it by recent sales misses by 92.9%.
- So the plans caught 21.6% of the 16,261 boxes that truly had to come down.
- 27 of 134 days needed more workers than planned.

People stay in charge

What its check raised	Days	
More workers than the roster	32	ASK
Demand the racks cannot cover	8	ASK
A SKU at 3 times its usual rate	76	CHECK
Weekly: SKU sizes not in the racks	33	CHECK
Weekly: idle boxes on levels 1-2	33	CHECK

The language model never calculates a number

Tools do every sum. With an Anthropic API key, Claude runs the same loop through the same tools, and a brief that states a number no tool produced is rejected. The scores here are the built-in rules'.

The records are real: 178 working days, 228,195 pairs, 208 SKUs and 41,256 boxes from one warehouse in Sherbrooke, Canada. They hold one snapshot of what sat where, so every morning starts from the same stock. No live warehouse is connected.

See it plan today

shreyasthakur11.github.io/warehouse-planning-agent

A live dashboard that shows a new plan each day: the brief, every step the agent took, and every plan checked against what happened. Nothing to install.

Data: de Assis, de Paula Ferreira and Ouhimmou (2025), CC BY 4.0